Pöschl–Teller Well in a Box: Full Derivations Behind the Summary

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How to use this document

Each section derives one part of `PT_Box_Summary.pdf`, in the same order:

Summary	What	Here
Assumptions box, "why αL"	A1–A6, dimensionless numbers	Sec. 1
Sec. 1, Eq. (1)	potential, source of its form	Sec. 1
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Every number is reproduced by `check_formulas.py` (about 10 s) and every figure by `make_figures.py`, both in this folder. The boxes marked **Check** list what each step relies on, so it can be verified line by line. Every citation gives the page (or the equation number) it refers to. Each bibliography entry lists all pages used and has a link to an online copy (and the local copy in `docs/refs`, where there is one). Sec. 8 says which copy each page number was checked against.

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1 The model and its two dimensionless numbers

A particle of mass m moves in one dimension with $H = p^2/2m + V(x)$, where

$$V(x) = \begin{cases} -\dfrac{\hbar^2}{2m}\,\alpha^2\,\dfrac{\lambda(\lambda-1)}{\cosh^2(\alpha x)}, & |x| < L/2, \\ +\infty, & |x| \geq L/2. \end{cases} \tag{1.1}$$

Inside the box this is the "modified Pöschl–Teller well" written exactly as in Table I of Deffner, Jarzynski and del Campo [1, p. 021013-9] (their q is our x). The table takes it from Flügge (their Ref. [55]). We take $\lambda = 2$, so $\lambda(\lambda-1) = 2$ and

$$V(x) = -\frac{\hbar^2}{2m}\,\alpha^2\,\frac{2}{\cosh^2(\alpha x)} = -\frac{\hbar^2\alpha^2}{m}\,\frac{1}{\cosh^2(\alpha x)} \qquad (|x| < L/2). \tag{1.2}$$

The well has depth $\hbar^2\alpha^2/m$ (classical floor $V_{\min} = -\hbar^2\alpha^2/m$ at $x = 0$) and width $\sim 1/\alpha$. For integer λ the well is *reflectionless*: its supersymmetric partner is the free particle (for $\lambda = 2$) or a shallower well of the same family, so the transmission amplitude has modulus 1 at every energy [2, p. 18, Eqs. (32)–(33); pp. 41–42, Eqs. (150)–(153)]. This is what makes the spectrum below exact.

Helpers. Only these derivation-local combinations are used, each defined in physical constants:

$$a \equiv \frac{\beta\hbar^2\alpha^2}{m} = \frac{\text{well depth}}{k_BT}, \qquad \lambda_{\text{th}} \equiv \sqrt{\frac{2\pi\hbar^2\beta}{m}}, \qquad \frac{L}{\lambda_{\text{th}}} = \frac{\alpha L}{\sqrt{2\pi a}}, \qquad \frac{k_BT}{\hbar^2\alpha^2/2m} = \frac{2}{a}. \tag{1.3}$$

The last two identities follow from $\lambda_{\text{th}}^2 = 2\pi\hbar^2\beta/m = 2\pi a/\alpha^2$. Every result below depends on αL and a only, i.e. on αL and $k_BT/(\hbar^2\alpha^2/2m)$. Temperatures are quoted as k_BT in units of $\hbar^2\alpha^2/2m$.

Why αL and not L. The constants of the problem are $\hbar$, m, α, L and k_BT. From $\hbar$, m and α one can build exactly one length, $1/\alpha$ (the well's width), and one energy, $\hbar^2\alpha^2/2m$. A length such as L can therefore enter a dimensionless quantity like C_v/k_B only as the ratio $L/(1/\alpha) = \alpha L$, the box length measured in well widths, and T only as $k_BT/(\hbar^2\alpha^2/2m)$. Two boxes with the same αL have identical C_v curves on this temperature axis, whatever L is in metres. The explicit formulas confirm this: L appears only in $L/\lambda_{\text{th}} = \alpha L/\sqrt{2\pi a}$ and in $\alpha L + \alpha I(a)$. It is also the quantity a scale-invariant stroke keeps fixed (Sec. 7).

The wall region. At the wall, $|V(\pm L/2)| = (\hbar^2\alpha^2/m)/\cosh^2(\alpha L/2) \approx 4(\hbar^2\alpha^2/m)\, e^{-\alpha L}$. Every approximation below drops terms of this order, which is below 10^{-40} of the depth for the cases used ($\alpha L \geq 100$).

2 Energy levels

2.1 Factorization: the well is the partner of the free particle

Following the supersymmetric method of Cooper, Khare and Sukhatme [2, pp. 13–14, Eqs. (3)–(11)], define the superpotential $W(x) = \frac{\hbar\alpha}{\sqrt{2m}}\tanh(\alpha x)$ and

$$A = \frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W(x), \qquad A^\dagger = -\frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W(x). \tag{2.1}$$

Acting on a test function f, $A^\dagger A f = -\frac{\hbar^2}{2m}f'' + \big(W^2 - \frac{\hbar}{\sqrt{2m}}W'\big)f$ and $AA^\dagger f = -\frac{\hbar^2}{2m}f'' + \big(W^2 + \frac{\hbar}{\sqrt{2m}}W'\big)f$. With $W^2 = \frac{\hbar^2\alpha^2}{2m}\tanh^2(\alpha x)$, $\frac{\hbar}{\sqrt{2m}}W' = \frac{\hbar^2\alpha^2}{2m}\frac{1}{\cosh^2(\alpha x)}$ and $\tanh^2 = 1 - 1/\cosh^2$:

$$A^\dagger A = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} - \frac{\hbar^2\alpha^2}{m}\frac{1}{\cosh^2(\alpha x)} + \frac{\hbar^2\alpha^2}{2m} = H_{\text{PT}} + \frac{\hbar^2\alpha^2}{2m}, \tag{2.2}$$

$$AA^\dagger = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{\hbar^2\alpha^2}{2m} = H_0 + \frac{\hbar^2\alpha^2}{2m}, \tag{2.3}$$

where H_{PT} has the potential of Eq. (1.2) and H_0 is the free particle. This is the pair $V_{1,2}$ of Ref. [2, p. 18, Eqs. (32)–(33)] with their $A = \hbar/\sqrt{2m}$ and $x \to \alpha x$. Multiplying $AA^\dagger = H_0 + \hbar^2\alpha^2/2m$ on the left by $A^\dagger$ and using the first line gives the intertwining relation $H_{\text{PT}}A^\dagger = A^\dagger H_0$. So if $H_0\phi = E\phi$ then $H_{\text{PT}}(A^\dagger\phi) = E(A^\dagger\phi)$. With $\phi = e^{ikx}$, $E = \hbar^2k^2/2m$, and dropping a constant factor,

$$\boxed{\psi_k(x) = \big(\alpha\tanh(\alpha x) - ik\big)\, e^{ikx}, \qquad E = \frac{\hbar^2k^2}{2m}.} \tag{2.4}$$

This solves the Schrödinger equation inside the box for every k, real or complex; no boundary condition has been used yet. Appendix A checks it by direct differentiation. (Ref. [2, p. 14, Eqs. (8)–(11)] states the same map between partner eigenfunctions.)

Check. (i) $\frac{\mathrm{d}}{\mathrm{d}x}\tanh(\alpha x) = \alpha/\cosh^2(\alpha x)$. (ii) Appendix A. (iii) Reflectionless: as $x \to \pm\infty$, $\psi_k \to (\pm\alpha - ik)e^{ikx}$, a pure right-moving wave on both sides, with no e^{-ikx} (reflected) part.

2.2 Solutions of definite parity

V is even, so the eigenstates have definite parity. For real k, the real and imaginary parts of Eq. (2.4) supply both:

$$\psi_o(x) = \mathrm{Re}\,\psi_k = \alpha\tanh(\alpha x)\cos(kx) + k\sin(kx) \qquad \text{(odd)}, \tag{2.5}$$

$$\psi_e(x) = -\mathrm{Im}\,\psi_k = k\cos(kx) - \alpha\tanh(\alpha x)\sin(kx) \qquad \text{(even)}. \tag{2.6}$$

2.3 Box levels ($E > 0$)

By parity, $\psi(\pm L/2) = 0$ reduces to one condition at $x = L/2$, where $\alpha\tanh(\alpha x) = \alpha\tanh(\alpha L/2) \equiv c$ (a derivation-local constant, $c = \alpha[1 + \mathcal{O}(e^{-\alpha L})]$):

$$\text{even:}\quad k\cos(kL/2) = c\,\sin(kL/2) \iff \tan(kL/2) = k/c, \tag{2.7}$$

$$\text{odd:}\quad c\cos(kL/2) + k\sin(kL/2) = 0 \iff \tan(kL/2) = -c/k. \tag{2.8}$$

Let $\varphi(k) = \arctan(k/c) \in (0, \pi/2)$. The even condition is $kL/2 = \varphi + \ell\pi$. Since $-c/k = -\cot\varphi = \tan(\varphi + \pi/2)$, the odd condition is $kL/2 = \varphi + \pi/2 + \ell\pi$. Together:

$$\boxed{k_j L = j\pi + 2\arctan\left(\frac{k_j}{\alpha\tanh(\alpha L/2)}\right), \qquad E_j = \frac{\hbar^2 k_j^2}{2m}, \qquad j = 1, 2, 3, \ldots} \tag{2.9}$$

with even j for even states and odd j for odd states. Dropping $\mathcal{O}(e^{-\alpha L})$, $\alpha\tanh(\alpha L/2) \to \alpha$, which is Eq. (2) of the summary.

Exactly one root per j. Let $f(k) = kL - 2\arctan(k/c)$. Then $f(0) = 0$ and $f'(k) = L - 2c/(k^2 + c^2) \geq L - 2/c > 0$ whenever $cL > 2$ (i.e. $\alpha L \gtrsim 2.4$). So f is increasing and each $j \geq 1$ has one root, in the bracket $j\pi/L < k_j < (j+1)\pi/L$ because $0 < 2\arctan < \pi$. The value $j = 0$ gives only $k = 0$, where $\psi_e \equiv 0$ and $\psi_o = \alpha\tanh(\alpha x)$ does not vanish at the wall, so it is not a state.

Solving it. The fixed-point iteration $k \leftarrow [j\pi + 2\arctan(k/c)]/L$, started at $k = j\pi/L$, contracts by a factor $\leq 2/(cL)$ per step (0.1 at $\alpha L = 20$). `check_formulas.py` reaches residuals of 10^{-15}.

2.4 The bound state ($E < 0$)

Set $k = i\kappa$, $\kappa > 0$, so $E = -\hbar^2\kappa^2/2m$. With $\cos(i\kappa x) = \cosh\kappa x$ and $\sin(i\kappa x) = i\sinh\kappa x$, Eqs. (2.5)–(2.6) give the real solutions

$$\chi_e = \kappa\cosh(\kappa x) - \alpha\tanh(\alpha x)\sinh(\kappa x), \qquad \chi_o = \alpha\tanh(\alpha x)\cosh(\kappa x) - \kappa\sinh(\kappa x). \tag{2.10}$$

- **Even:** $\chi_e(L/2) = 0 \iff$

$$\kappa_b = \alpha \tanh(\alpha L/2) \tanh(\kappa_b L/2), \qquad E_b = -\frac{\hbar^2 \kappa_b^2}{2m} = -\frac{\hbar^2 \alpha^2}{2m} \left[1 - 8e^{-\alpha L} + \dots\right]. \tag{2.11}$$

  $g(\kappa) = c \tanh(\kappa L/2) - \kappa$ has $g(0) = 0$, $g'(0) = cL/2 - 1 > 0$ and is concave, so there is exactly one root, with $\kappa_b < c < \alpha$. At $\alpha L = 20$, $E_b/(\hbar^2\alpha^2/2m) = -0.99999998351$, matching $1 - 8e^{-20}$. The wavefunction is $\chi_e \to \alpha/\cosh(\alpha x)$ as $\kappa \to \alpha$ (use $\cosh(\alpha x) - \tanh(\alpha x)\sinh(\alpha x) = 1/\cosh(\alpha x)$).

- **Odd:** $\chi_o(L/2) = 0 \iff \kappa \tanh(\kappa L/2) = c = \alpha \tanh(\alpha L/2)$. The left side increases with κ, so the only root is exactly $\kappa = \alpha$. But there $\chi_o \equiv 0$, because $\alpha \tanh(\alpha x)\cosh(\alpha x) = \alpha \sinh(\alpha x)$, so the root is spurious. At $E = -\hbar^2\alpha^2/2m$ the second (odd) solution is $[x/2 + \sinh(2\alpha x)/4\alpha]/\cosh(\alpha x)$, positive for all $x > 0$, so it cannot vanish at the wall. Hence there is *no odd bound state*.

Why exactly one bound state. This follows from $\lambda = 2$; it is not an extra assumption. Three independent arguments give it:

(i) *In the box*, which is the system studied here: the even condition has exactly one root and the odd condition has none (above). This holds exactly, for any $\alpha L \tanh(\alpha L/2) > 2$.

(ii) *On the open line, by supersymmetry.* $A^\dagger A = H_{\rm PT} + \hbar^2\alpha^2/2m$ and $AA^\dagger = H_0 + \hbar^2\alpha^2/2m$ have identical spectra, except that $A^\dagger A$ has one extra level at zero, the zero mode of A [2, p. 15, Eqs. (12)–(14)]. Equivalently, the partner "has one fewer bound state" [2, p. 17]. The partner here is the free particle, which has no bound state, so the well has exactly one. It is the zero mode, $A\psi_b = 0$, which gives $\psi_b \propto 1/\cosh(\alpha x)$ at $E_b = -\hbar^2\alpha^2/2m$.

(iii) *The literature value for this family.* For $V_1 = -n(n+1)\,\mathrm{sech}^2 x$ (units $\hbar = 2m = \alpha = 1$), Cooper, Khare and Sukhatme state that "V_1 holds n bound states" [2, p. 77, Eq. (312)]. Matching $n(n+1) = \lambda(\lambda-1) = 2$ gives $n = \lambda - 1 = 1$. An integer λ gives $\lambda - 1$ bound states, so $\lambda = 2$ is the smallest reflectionless well that binds at all ($\lambda = 1$ has no well).

Check against the literature. Cooper, Khare and Sukhatme [2, pp. 77–78] treat exactly this well on the open line, in units $\hbar = 2m = 1$ and $\alpha = 1$: "$V_1 = -2\,\mathrm{sech}^2 x$, $E_1 = -1$", with ground state $\propto \mathrm{sech}\, x$ (their case $n = 1$). Restoring units, $V_1 = -\frac{\hbar^2\alpha^2}{2m}\frac{2}{\cosh^2(\alpha x)}$ is Eq. (1.2) and $E_1 = -\hbar^2\alpha^2/2m$ is Eq. (2.11) as $\alpha L \to \infty$. Independently, $A^\dagger A = H_{\rm PT} + \hbar^2\alpha^2/2m$ annihilates $1/\cosh(\alpha x)$ (since $A\frac{1}{\cosh \alpha x} = 0$), which gives the same E_b.

2.5 Physical reading: phase shift and Levinson's theorem

For the empty box $k_j L = j\pi$. The well adds $2\arctan(k/\alpha) \in (0, \pi)$, so every box level is pushed up, by less than one spacing (Table 1). As $k \to \infty$ the addition tends to π, so level j approaches level $j + 1$ of the empty box: one state has left the continuum, and it reappears as the bound state.

In the standard convention the scattering phase shift is $\delta(k) = \pi - 2\arctan(k/\alpha) = 2\arctan(\alpha/k)$, and Eq. (2.9) reads $k_j L = (j+1)\pi - \delta(k_j)$. It falls from $\delta(0) = \pi$ to $\delta(\infty) = 0$, so $\delta(0) - \delta(\infty) = \pi \times$ (one bound state), the statement known as Levinson's theorem. Here it needs no separate proof: the counting above shows it directly from the bracket $j\pi/L < k_j < (j+1)\pi/L$. The level density used in Sec. 3 is $\mathrm{d}j/\mathrm{d}k = [L + \mathrm{d}\delta/\mathrm{d}k]/\pi$.

Table 1: The levels drawn in the summary's Fig. 1(a), $\alpha L = 20$, in units of $\hbar^2\alpha^2/2m$ (`check_formulas.py`, section 1).

	bound	$j=1$	$j=2$	$j=3$	$j=4$
PT + box, Eqs. (2.9)–(2.11)	−0.99999998	0.030394	0.120836	0.269487	0.474202
empty box, $(j\pi/\alpha L)^2$	–	0.024674	0.098696	0.222066	0.394784

3 Quantum partition function and heat capacity

3.1 The variance formula

For levels $\{E_n\}$, $Z = \sum_n e^{-\beta E_n}$, $\langle E\rangle = -\partial_\beta \ln Z$ and $\langle E^2\rangle = Z^{-1}\partial_\beta^2 Z$. With $\partial_T = -k_B\beta^2\partial_\beta$,

$$C_v = \frac{\partial\langle E\rangle}{\partial T} = k_B\beta^2\frac{\partial^2 \ln Z}{\partial\beta^2} = k_B\beta^2\left(\langle E^2\rangle - \langle E\rangle^2\right) = k_B\beta^2\,\mathrm{Var}(E), \tag{3.1}$$

the standard energy-fluctuation result of the canonical ensemble, $\Delta E^2 = k_B T^2 C_V$ [3, p. 20, §1.3.2, Eq. (1.27)]. Since $a = \beta\hbar^2\alpha^2/m$ is proportional to β, $\beta\,\partial_\beta = a\,\partial_a$ and $\beta^2\partial_\beta^2 = a^2\partial_a^2$, so equivalently $C_v = k_B\,a^2\,\partial_a^2 \ln Z$. Here

$$Z_q = e^{-\beta E_b} + \sum_{j=1}^{\infty} e^{-\beta E_j}. \tag{3.2}$$

3.2 The sum over box levels by Poisson summation

Let $j(k) = [kL - 2\arctan(k/c)]/\pi$ be the counting function implied by Eq. (2.9), $k(j)$ its inverse, and $F(j) = \exp\!\left[-\beta\hbar^2 k(j)^2/2m\right]$, so the box sum is $\sum_{j\ge1} F(j)$. Since arctan is odd, $j(k)$ and $k(j)$ are odd, so F is even with $F(0) = 1$. Poisson's summation formula in the form of DLMF Eq. (1.8.15) [5],

$$\tfrac{1}{2}F(0) + \sum_{n=1}^{\infty} F(n) = \int_0^\infty F(y)\,\mathrm{d}y + 2\sum_{\ell=1}^{\infty}\int_0^\infty F(y)\cos(2\pi\ell y)\,\mathrm{d}y,$$

gives

$$\sum_{j=1}^{\infty} F(j) = \int_0^\infty F(j)\,\mathrm{d}j - \frac{1}{2} + R, \qquad R = 2\sum_{\ell\ge1}\int_0^\infty F(y)\cos(2\pi\ell y)\,\mathrm{d}y. \tag{3.3}$$

R is exponentially small. For the empty box, $F(y) = e^{-\pi(\lambda_{\mathrm{th}}y/2L)^2}$ and each term is $(2L/\lambda_{\mathrm{th}})\,e^{-4\pi\ell^2(L/\lambda_{\mathrm{th}})^2}$ (the cosine transform of a Gaussian); the well only adds terms small in $e^{-\alpha L}$. It is below double precision whenever $L \gtrsim \lambda_{\mathrm{th}}$ (Sec. 5 shows where it starts to matter).

3.3 The integral, using the level density

Change variables from j to k. The level density, read off Eq. (2.9), is

$$\frac{\mathrm{d}j}{\mathrm{d}k} = \frac{1}{\pi}\left[L - \frac{2c}{k^2+c^2}\right]. \tag{3.4}$$

The first term is the empty box; the second, $-\frac{1}{\pi}\frac{\mathrm{d}}{\mathrm{d}k}[2\arctan(k/c)] = \frac{1}{\pi}\frac{\mathrm{d}\delta}{\mathrm{d}k}$, is the change in the density of states caused by the phase shift. With the derivation-local $b = \beta\hbar^2/2m$ (so $\lambda_{\mathrm{th}} = \sqrt{4\pi b}$):

$$\begin{aligned}\int_0^\infty F\,\mathrm{d}j &= \frac{L}{\pi}\int_0^\infty e^{-bk^2}\mathrm{d}k - \frac{2c}{\pi}\int_0^\infty \frac{e^{-bk^2}}{k^2+c^2}\,\mathrm{d}k \\ &= \frac{L}{\pi}\cdot\frac{1}{2}\sqrt{\frac{\pi}{b}} - \frac{2c}{\pi}\cdot\frac{\pi}{2c}\,e^{bc^2}\operatorname{erfc}\!\left(c\sqrt{b}\right) = \frac{L}{\lambda_{\mathrm{th}}} - e^{bc^2}\operatorname{erfc}\!\left(c\sqrt{b}\right). \end{aligned} \tag{3.5}$$

The second integral is DLMF Eq. (7.7.1) [5], $\operatorname{erfc} z = \frac{2}{\pi}e^{-z^2}\int_0^\infty \frac{e^{-z^2t^2}}{t^2+1}\mathrm{d}t$, with $k = ct$ and $z = c\sqrt{b}$; Appendix B derives it independently.

3.4 The closed form

Add the bound state, $e^{-\beta E_b} = e^{\beta\hbar^2\kappa_b^2/2m}$:

$$Z_q = \frac{L}{\lambda_{\mathrm{th}}} - \frac{1}{2} + e^{\beta\hbar^2\kappa_b^2/2m} - e^{bc^2}\operatorname{erfc}\left(c\sqrt{b}\right) + R. \tag{3.6}$$

Both κ_b and c equal $\alpha[1+\mathcal{O}(e^{-\alpha L})]$, and $b\alpha^2 = \beta\hbar^2\alpha^2/2m = a/2$. Using $1-\operatorname{erfc} = \operatorname{erf}$:

$$\boxed{Z_q = \frac{L}{\lambda_{\mathrm{th}}} - \frac{1}{2} + e^{a/2}\operatorname{erf}\sqrt{a/2}\,, \qquad a = \frac{\beta\hbar^2\alpha^2}{m}.} \tag{3.7}$$

Reading. L/λ_{th}: free particle in the box. $-\frac{1}{2}$: hard-wall (Dirichlet) correction, half a state missing. $e^{a/2} = e^{-\beta E_b}$: the bound state. $-e^{a/2}\operatorname{erfc}\sqrt{a/2}$: the continuum state that became the bound state, removed with the energy spread set by the phase shift, i.e. $\frac{1}{\pi}\int_0^\infty \frac{\mathrm{d}\delta}{\mathrm{d}k}e^{-\beta\hbar^2k^2/2m}\mathrm{d}k$.

Check. (i) $\frac{L}{\pi}\cdot\frac{1}{2}\sqrt{\pi/b} = L/\sqrt{4\pi b} = L/\lambda_{\mathrm{th}}$. (ii) Appendix B. (iii) Numerically, Eq. (3.7) with Eq. (3.10) below against $\mathrm{Var}(E)$ over the exact levels of Eqs. (2.9)–(2.11): max difference $1.1\times10^{-13}k_B$, $2.2\times10^{-13}k_B$ and $2.1\times10^{-13}k_B$ for $\alpha L = 60$, 100 and 1000, over $k_BT \in [0.02, 5]\,(\hbar^2\alpha^2/2m)$, using 381, 602 and 5564 levels.

3.5 Closed-form C_v^{q}

Write $Z_q = B(a) - \frac{1}{2} + G(a)$ with $B = L/\lambda_{\mathrm{th}} = \alpha L/\sqrt{2\pi a}$ and $G = e^{a/2}\operatorname{erf}\sqrt{a/2}$. Primes are $\mathrm{d}/\mathrm{d}a$. Since $B \propto a^{-1/2}$,

$$B' = -\frac{B}{2a}, \qquad B'' = \frac{3B}{4a^2}. \tag{3.8}$$

From $\operatorname{erf} z = \frac{2}{\sqrt{\pi}}\int_0^z e^{-s^2}\mathrm{d}s$ (DLMF Eq. (7.2.1) [5]), $\frac{\mathrm{d}}{\mathrm{d}a}\operatorname{erf}\sqrt{a/2} = \frac{2}{\sqrt{\pi}}e^{-a/2}\cdot\frac{1}{2\sqrt{2a}} = \frac{e^{-a/2}}{\sqrt{2\pi a}}$, so

$$G' = \frac{1}{2}G + \frac{1}{\sqrt{2\pi a}}, \qquad G'' = \frac{1}{2}G' - \frac{1}{2\sqrt{2\pi}}\,a^{-3/2}. \tag{3.9}$$

Then $Z_q' = B' + G'$, $Z_q'' = B'' + G''$, and Eq. (3.1) becomes

$$\boxed{\frac{C_v^{\mathrm{q}}}{k_B} = a^2\left[\frac{Z_q''}{Z_q} - \left(\frac{Z_q'}{Z_q}\right)^2\right].} \tag{3.10}$$

(In `cv_plot.py`, Z_q, Z_q' and Z_q'' are all multiplied by $e^{-a/2}$ before dividing. The ratios are unchanged and nothing overflows at low T.)

3.6 Limits

- $T \to 0$ ($a \to \infty$): $Z_q \to e^{a/2}$ and $C_v^{\mathrm{q}} \to 0$ exponentially, because of the gap $\hbar^2\alpha^2/2m$ between the bound state and the bottom of the continuum.
- $T \to \infty$ ($a \to 0$): $G = e^{a/2}\operatorname{erf}\sqrt{a/2} \simeq \sqrt{2a/\pi}$, which equals the classical well term I/λ_{th} at the same order (Sec. 4: $\alpha I \simeq 2a$ and $\alpha\lambda_{\mathrm{th}} = \sqrt{2\pi a}$). Only the $-\frac{1}{2}$ is left to distinguish Z_q from Z_{cl}, so $\ln Z_q - \ln Z_{\mathrm{cl}} \simeq -\frac{\lambda_{\mathrm{th}}}{2L} \propto \beta^{1/2}$. With $\beta^2\partial_\beta^2\beta^{1/2} = -\frac{1}{4}\beta^{1/2}$,

$$C_v^{\mathrm{q}} - C_v^{\mathrm{cl}} \xrightarrow{T\to\infty} k_B\,\frac{\lambda_{\mathrm{th}}}{8L} > 0, \tag{3.11}$$

  which is the wall effect of Sec. 5. Numerically, at $k_BT = 1000\,(\hbar^2\alpha^2/2m)$ and $\alpha L = 100$: $C_v^{\mathrm{q}} = 0.50014k_B$, $C_v^{\mathrm{cl}} = 0.50000k_B$, $\lambda_{\mathrm{th}}/8L = 0.00014$.

4 Classical partition function and heat capacity

4.1 Phase-space integral

The classical canonical partition function of one particle is $Z_1 = \frac{1}{h^3}\int \mathrm{d}^3q\,\mathrm{d}^3p\,e^{-\beta H}$ [3, p. 32, §2.1, Eq. (2.1)]; in one dimension, with $h = 2\pi\hbar$ and the thermal wavelength of Ref. [3, p. 35, Eq. (2.6)],

$$Z_{\mathrm{cl}} = \frac{1}{2\pi\hbar}\int_{-L/2}^{L/2}\mathrm{d}x\int_{-\infty}^{\infty}\mathrm{d}p\,e^{-\beta H} = \frac{1}{2\pi\hbar}\sqrt{\frac{2\pi m}{\beta}}\int_{-L/2}^{L/2}e^{-\beta V}\mathrm{d}x = \frac{1}{\lambda_{\mathrm{th}}}\int_{-L/2}^{L/2}e^{-\beta V(x)}\mathrm{d}x. \tag{4.1}$$

Split off the flat part of the box: $\int e^{-\beta V}\mathrm{d}x = L + \int(e^{-\beta V}-1)\mathrm{d}x$, and extend the second integral to $\mathbb{R}$, which costs $\mathcal{O}(e^{-\alpha L})$. Then

$$Z_{\mathrm{cl}} = \frac{L+I}{\lambda_{\mathrm{th}}}, \qquad I = \int_{-\infty}^{\infty}\left(e^{-\beta V}-1\right)\mathrm{d}x, \qquad -\beta V = \frac{a}{\cosh^2(\alpha x)}. \tag{4.2}$$

4.2 The well integral I

Substitute $t = \tanh(\alpha x)$: $1/\cosh^2(\alpha x) = 1 - t^2$ and $\mathrm{d}x = \mathrm{d}t/[\alpha(1-t^2)]$. Then

$$\alpha I = \int_{-1}^{1}\frac{e^{a(1-t^2)}-1}{1-t^2}\,\mathrm{d}t = \sum_{\ell\ge1}\frac{a^\ell}{\ell!}\int_{-1}^{1}(1-t^2)^{\ell-1}\mathrm{d}t = \sum_{\ell\ge1}\frac{a^\ell}{\ell!}\,B\!\left(\tfrac{1}{2},\ell\right) = \sum_{\ell\ge1}\frac{(2a)^\ell}{\ell\,(2\ell-1)!!}. \tag{4.3}$$

Here $\int_{-1}^{1}(1-t^2)^{\ell-1}\mathrm{d}t = \int_0^1 s^{-1/2}(1-s)^{\ell-1}\mathrm{d}s = B(\frac{1}{2},\ell) = \Gamma(\frac{1}{2})\Gamma(\ell)/\Gamma(\ell+\frac{1}{2})$ (with $s = t^2$; DLMF Eq. (5.12.1) [5]). Applying $\Gamma(z+1) = z\Gamma(z)$ (DLMF Eq. (5.5.1)) ℓ times, $\Gamma(\ell+\frac{1}{2}) = \frac{(2\ell-1)!!}{2^\ell}\Gamma(\frac{1}{2})$, so $B(\frac{1}{2},\ell) = 2^\ell(\ell-1)!/(2\ell-1)!!$ (with $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, DLMF Eq. (5.4.6), which cancels). The series is $2a\,{}_2F_2(1,1;\frac{3}{2},2;a)$ and converges for every a. `cv_plot.py` evaluates the integral form by Gauss–Legendre quadrature, which agrees with the series to 1.2×10^{-13} (relative).

4.3 Derivatives and closed-form C_v^{cl}

Up to a constant, $\ln Z_{\mathrm{cl}} = -\frac{1}{2}\ln a + \ln(\alpha L + \alpha I)$. Differentiating Eq. (4.3) under the integral,

$$\alpha J_1 \equiv \frac{\mathrm{d}(\alpha I)}{\mathrm{d}a} = \int_{-1}^{1}e^{a(1-t^2)}\mathrm{d}t = 2e^a\int_0^1 e^{-at^2}\mathrm{d}t = \sqrt{\pi}\,\frac{e^a\operatorname{erf}\sqrt{a}}{\sqrt{a}}, \tag{4.4}$$

$$\alpha J_2 \equiv \frac{\mathrm{d}(\alpha J_1)}{\mathrm{d}a} = \int_{-1}^{1}(1-t^2)\,e^{a(1-t^2)}\mathrm{d}t = \alpha J_1\left(1-\frac{1}{2a}\right)+\frac{1}{a}, \tag{4.5}$$

where the last step uses $\mathrm{d}\,\mathrm{erf}\sqrt{a}/\mathrm{d}a = e^{-a}/\sqrt{\pi a}$ (Appendix C). In terms of x,

$$J_k = \int \frac{1}{\cosh^{2k}(\alpha x)} \exp\!\left(\frac{a}{\cosh^2(\alpha x)}\right) \mathrm{d}x.$$

Since $a^2\partial_a^2(-\frac{1}{2}\ln a) = \frac{1}{2}$,

$$\boxed{\frac{C_v^{\mathrm{cl}}}{k_B} = \frac{1}{2} + a^2\left[\frac{\alpha J_2}{\alpha L + \alpha I} - \left(\frac{\alpha J_1}{\alpha L + \alpha I}\right)^2\right].} \tag{4.6}$$

The physical reading. The $\frac{1}{2}$ is the kinetic energy (equipartition, $p^2/2m$). The bracket is $a^2\,\mathrm{Var}\big(1/\cosh^2\alpha x\big) = \beta^2\,\mathrm{Var}(V)$, the variance of the potential energy under the Boltzmann distribution $P(x) \propto e^{-\beta V(x)}$ in the box. This form is what `check_formulas.py` integrates directly, to check Eq. (4.6): max difference $1.1 \times 10^{-9} k_B$.

4.4 Limits

- $T \to 0$: the particle sits in the harmonic bottom, $V \simeq -\frac{\hbar^2\alpha^2}{m} + \frac{\hbar^2\alpha^4}{m}x^2$, and equipartition over x^2 and p^2 gives $C_v^{\mathrm{cl}} \to k_B$, the classical oscillator value [6, p. 95]. Numerically $C_v^{\mathrm{cl}} = 1.0010 k_B$ at $k_B T = 0.002\,(\hbar^2\alpha^2/2m)$; the excess is the anharmonic correction.
- $T \to \infty$: the well is irrelevant and $C_v^{\mathrm{cl}} \to k_B/2$.

5 The empty box: the wall effect

Remove the well. The levels are $E_n = n^2 E_1$, $E_1 = \pi^2\hbar^2/2mL^2$, and Sec. 3.2 with no phase shift gives

$$Z_{\mathrm{box}} = \sum_{n\geq 1} e^{-\beta E_1 n^2} = \frac{L}{\lambda_{\mathrm{th}}} - \frac{1}{2} + \frac{2L}{\lambda_{\mathrm{th}}}\sum_{\ell\geq 1} e^{-4\pi\ell^2(L/\lambda_{\mathrm{th}})^2}. \tag{5.1}$$

Classically $Z_{\mathrm{cl}} = L/\lambda_{\mathrm{th}}$ exactly, so $C_v^{\mathrm{cl}} = k_B/2$ at every T. Drop the exponentially small remainder and write $y = L/\lambda_{\mathrm{th}} \propto \beta^{-1/2}$, so $\beta y' = -y/2$ and $\beta^2 y'' = 3y/4$. With $Z = y - \frac{1}{2}$,

$$\frac{C_v^{\mathrm{q}}}{k_B} = \beta^2\left[\frac{Z''}{Z} - \left(\frac{Z'}{Z}\right)^2\right] = \frac{\frac{3}{4}y\,(y-\frac{1}{2}) - \frac{1}{4}y^2}{(y-\frac{1}{2})^2} = \frac{y(2y-\frac{3}{2})}{4(y-\frac{1}{2})^2}, \tag{5.2}$$

and subtracting $\frac{1}{2}$,

$$\boxed{\frac{C_v^{\mathrm{q}} - C_v^{\mathrm{cl}}}{k_B} = \frac{y-1}{8\,(y-\frac{1}{2})^2} = \frac{r(1-r)}{2(2-r)^2}, \qquad r = \frac{\lambda_{\mathrm{th}}}{L} = \frac{1}{y}.} \tag{5.3}$$

- It is positive iff $r < 1$, i.e. $L > \lambda_{\mathrm{th}}$, i.e. $k_B T > \frac{4}{\pi}E_1$.
- $\frac{\mathrm{d}}{\mathrm{d}r}\frac{r(1-r)}{(2-r)^2} = \frac{2(1-\frac{3}{2}r)}{(2-r)^3}$ vanishes at $r = \frac{2}{3}$. The maximum is $\frac{(2/3)(1/3)}{2(4/3)^2} = \frac{1}{16}$, at $L = \frac{3}{2}\lambda_{\mathrm{th}}$, i.e. $k_B T = \frac{9}{\pi}E_1 = 2.865 E_1$.
- As $T \to \infty$ it tends to $r/8 = \lambda_{\mathrm{th}}/8L$, the same wall term as Eq. (3.11).

Heuristic. $Z = (L - \lambda_{\mathrm{th}}/2)/\lambda_{\mathrm{th}}$: the hard walls make a layer of thickness $\sim \lambda_{\mathrm{th}}/4$ at each wall unavailable. Heating thins the layer ($\lambda_{\mathrm{th}} \propto T^{-1/2}$) and returns space to the particle, and that costs extra heat. Levy and Gelbwaser-Klimovsky show the same box result in their Fig. 4.3 [6, p. 95; discussed on p. 96].

Validity and check (Fig. 1). Against the exact level sum, Eq. (5.3) is good to 2×10^{-3} for $r < 1$, 10^{-9} for $r < \frac{2}{3}$ and 10^{-16} for $r < \frac{1}{2}$. The exact sum peaks at $0.562500 k_B$ at $k_B T = 2.864 E_1$ and exceeds $k_B/2$ from $k_B T = 1.265 E_1$ (the closed form says $4/\pi = 1.273$).

Size relative to the well's anomaly. In the anomaly windows of Sec. 6, $r = \lambda_{\mathrm{th}}/L$ is small. At $\alpha L = 100$, $k_B T = 0.21\,(\hbar^2\alpha^2/2m)$, $r = 0.077$ and Eq. (5.3) gives $0.0097 k_B$, against the total excess $0.34 k_B$. At $\alpha L = 1000$ it is about $10^{-3} k_B$, against $4.8 k_B$.

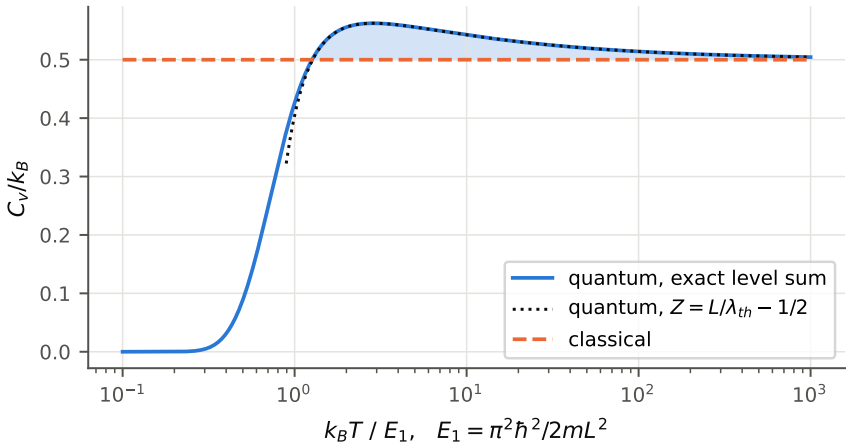


Figure 1: Empty box: exact level sum (blue), Eq. (5.3) (dotted, drawn for $r < 1.2$) and the classical $k_B/2$ (orange, dashed). The curve is the same for every L when T is measured in units of E_1.

6 The mechanism, and the numbers

6.1 Results

Table 2 and Fig. 2 compare Eqs. (3.10) and (4.6).

Table 2: Quantum vs. classical heat capacity (`check_formulas.py`, section 4). All temperatures are $k_B T$ in units of $\hbar^2\alpha^2/2m$, on a grid up to 1. "Window" is where $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$.

αL	$\max(C_v^{\mathrm{q}} - C_v^{\mathrm{cl}})/k_B$	at $k_B T$	C_v^{q}/k_B, C_v^{cl}/k_B there	window	C_v^{q} peak (at)	C_v^{cl} peak (at)	$\min(C_v^{\mathrm{q}} - C_v^{\mathrm{cl}})/k_B$ (at)
30	none for $k_B T < 1$	–	–	–	2.00 (0.393)	3.74 (0.412)	−1.75 (0.421)
100	+0.343	0.210	2.427, 2.084	[0.176, 0.239]	3.56 (0.285)	6.21 (0.344)	−3.18 (0.367)
400	+2.591	0.184	5.091, 2.499	[0.129, 0.223]	6.05 (0.216)	10.03 (0.283)	−6.20 (0.299)
1000	+4.802	0.167	7.397, 2.595	[0.112, 0.203]	8.13 (0.185)	13.13 (0.252)	−9.10 (0.263)

6.2 Heuristic 1: who desorbs first

Both curves are dominated by one event, the particle leaving the well for the box (desorption). The transition sits roughly where the well and the box carry equal Boltzmann weight.

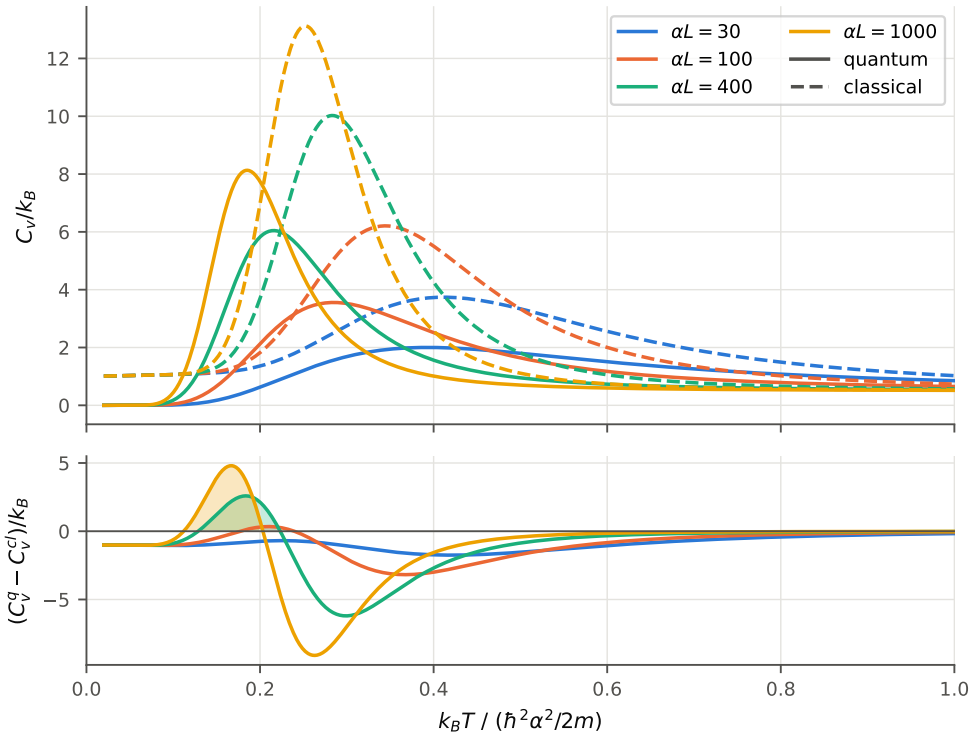


Figure 2: C_v^{q} (solid) and C_v^{cl} (dashed) and their difference, for four box lengths. Shaded: $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$.

- **Quantum.** The well's weight is the single bound state, $e^{-\beta E_b} = e^{a/2}$; the box's is L/λ_{th}. Equal weights give, with Eq. (1.3),

$$\frac{a_q}{2} = \ln\frac{L}{\lambda_{\mathrm{th}}} \quad \Longleftrightarrow \quad \frac{\hbar^2\alpha^2/2m}{k_B T_q} = \ln\left(\alpha L\sqrt{\frac{k_B T_q}{4\pi\,(\hbar^2\alpha^2/2m)}}\right). \tag{6.1}$$

- **Classical.** The well's weight is I/λ_{th}. At low T only the harmonic bottom matters, and Laplace's method gives $\alpha I \simeq e^a\sqrt{\pi/a}$, i.e. $I/\lambda_{\mathrm{th}} \simeq e^a\, k_B T/\hbar\omega_0$ with $\hbar\omega_0 = \sqrt{2}\,\hbar^2\alpha^2/m$, the frequency of the bottom ($\frac{1}{2}m\omega_0^2 = \hbar^2\alpha^4/m$). The factor $k_B T/\hbar\omega_0$ is the classical oscillator's partition function: the entropy cost of being localized. Equal weights, $I = L$, give

$$a_{\mathrm{cl}} - \tfrac{1}{2}\ln a_{\mathrm{cl}} = \ln\frac{\alpha L}{\sqrt{\pi}}, \qquad \frac{k_B T_{\mathrm{cl}}}{\hbar^2\alpha^2/2m} = \frac{2}{a_{\mathrm{cl}}}. \tag{6.2}$$

In the relevant range the classical well holds more weight than the quantum one: their ratio is $[e^a/(\sqrt{2}a)]/e^{a/2} = e^{a/2}/(\sqrt{2}a)$, about 10 at $a = 10$. The zero-point energy costs the quantum particle half the depth, more than localization costs the classical one. Hence $T_q < T_{\mathrm{cl}}$:

αL	$k_B T_q$ estimate	C_v^{q} peak	$k_B T_{\mathrm{cl}}$ estimate	C_v^{cl} peak
100	0.354	0.285	0.415	0.344
400	0.248	0.216	0.315	0.283
1000	0.206	0.185	0.273	0.252

The estimates run 10–25% high but reproduce the order $T_q < T_{\mathrm{cl}}$ and the growth of the gap with αL.

6.3 Heuristic 2: the peak heights

Treat each system as two "phases", bound and gas, with occupation p of the gas. A Schottky-type estimate gives a peak $C_v \approx k_B\, p(1-p)(\beta\Delta)^2 \le \frac{k_B}{4}(\beta\Delta)^2$, where Δ is the energy gained on desorption. Quantum: from $-\hbar^2\alpha^2/2m$ to a gas with mean energy $\frac{1}{2}k_BT$, so $\Delta_q = \hbar^2\alpha^2/2m + \frac{1}{2}k_BT$. Classical: from $-\hbar^2\alpha^2/m + k_BT$ (harmonic bottom) to $\frac{1}{2}k_BT$, so $\Delta_{\mathrm{cl}} = \hbar^2\alpha^2/m - \frac{1}{2}k_BT$. In units of $\hbar^2\alpha^2/2m$, $\beta\Delta_q = \hbar^2\alpha^2/2m/k_BT + \frac{1}{2}$ and $\beta\Delta_{\mathrm{cl}} = 2\,\hbar^2\alpha^2/2m/k_BT - \frac{1}{2}$. At the computed peak temperatures this gives 4.0, 6.6, 8.7 (quantum) and 7.1, 10.8, 13.8 (classical) for $\alpha L = 100$, 400, 1000, against the computed 3.56, 6.05, 8.13 and 6.21, 10.03, 13.13, i.e. 5–15% high. The classical peak is taller because its latent heat is about twice as large; the quantum peak comes first.

6.4 The window

Below both peaks, $C_v^{\mathrm{q}} \approx 0$ (frozen in one state) while $C_v^{\mathrm{cl}} \approx k_B$ (vibrating in the bottom), so $C_v^{\mathrm{q}} < C_v^{\mathrm{cl}}$. As T rises the quantum particle starts to desorb first. Its C_v^{q} rises steeply while the classical particle is still bound with $C_v^{\mathrm{cl}} \approx k_B$, and this is the window where $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$. Then the classical desorption begins, with its larger latent heat, and the order reverses into a deficit (last column of Table 2). The anomaly is a *shift* of the peak to lower temperature, not a larger peak. A bigger box (larger αL) means more states to desorb into, lower desorption temperatures, larger $\beta\Delta$ and sharper peaks, so the excess grows with αL. The window first opens between $\alpha L = 70$ and 80: $\max(C_v^{\mathrm{q}} - C_v^{\mathrm{cl}}) = -0.04k_B$ at $\alpha L = 70$ and $+0.10k_B$ at $\alpha L = 80$.

7 Homogeneous (scale-invariant) strokes

7.1 Definition

Deffner *et al.* [1, p. 021013-2, Eq. (3)] call a driving scale-invariant when the Hamiltonian keeps the form of their Eq. (3). For a dilation about $x = 0$ (their $f = 0$; their U and q are our V and x),

$$H(\gamma) = \frac{p^2}{2m} + \frac{1}{\gamma^2}\,V_0\!\left(\frac{x}{\gamma}\right). \tag{7.1}$$

A family $V(x;\mu)$ with control parameters μ admits such a stroke if there is a path $\mu(\gamma)$ with $\mu(1) = \mu_0$ and

$$V\big(x;\mu(\gamma)\big) = \frac{1}{\gamma^2}\,V\!\left(\frac{x}{\gamma};\mu_0\right) \qquad \text{for all } x \text{ and all } \gamma > 0. \tag{7.2}$$

7.2 Each component, and their sum

Write $V(x;\alpha,L) = V_{\mathrm{PT}}(x;\alpha) + V_{\mathrm{box}}(x;L)$ with $V_{\mathrm{PT}} = -\frac{\hbar^2}{2m}\alpha^2\frac{\lambda(\lambda-1)}{\cosh^2(\alpha x)}$ on $\mathbb{R}$ and $V_{\mathrm{box}} = 0$ for $|x| < L/2$, $+\infty$ otherwise.

- **Well** (Table I of Ref. [1, p. 021013-9]: $\alpha \to \alpha/\gamma$, λ fixed):

$$V_{\mathrm{PT}}\!\left(x;\tfrac{\alpha}{\gamma}\right) = -\frac{\hbar^2}{2m}\frac{\alpha^2}{\gamma^2}\frac{\lambda(\lambda-1)}{\cosh^2(\alpha x/\gamma)} = \frac{1}{\gamma^2}V_{\mathrm{PT}}\!\left(\frac{x}{\gamma};\alpha\right).$$

  The α^2 in front supplies γ^{-2}; the α inside cosh supplies x/γ.

- **Box** ($L \to \gamma L$; Table I lists it as the $b \to \infty$ member of the power-law family): $|x| < \gamma L/2 \iff |x/\gamma| < L/2$, and $\gamma^{-2}\cdot 0 = 0$, $\gamma^{-2}\cdot\infty = \infty$, so $V_{\mathrm{box}}(x;\gamma L) = \gamma^{-2}V_{\mathrm{box}}(x/\gamma;L)$.

- **Sum, same γ.** The map $U \mapsto \gamma^{-2}U(x/\gamma)$ is linear, so

$$V\left(x; \frac{\alpha}{\gamma}, \gamma L\right) = \frac{1}{\gamma^2} V\left(\frac{x}{\gamma}; \alpha, L\right). \tag{7.3}$$

- **Different factors fail.** With $\alpha \to \alpha/\gamma_1$ and $L \to \gamma_2 L$, suppose the result were $c^{-2}V(x/c; \alpha, L)$. The walls force $c = \gamma_2$; the value at $x = 0$ forces $\alpha^2/\gamma_1^2 = \alpha^2/c^2$, i.e. $c = \gamma_1$. So $\gamma_1 = \gamma_2$. Physically: the shape is set by αL, and a scale-invariant stroke must keep it.

7.3 All levels scale by the same factor

Let ψ_n solve the $\gamma = 1$ problem with energy E_n and $\psi_n(\pm L/2) = 0$, and define $\psi_n^{(\gamma)}(x) = \gamma^{-1/2}\psi_n(x/\gamma)$ (the map of Ref. [1, p. 021013-14, App. A]). Since $\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi_n(x/\gamma) = \gamma^{-2}\psi_n''(x/\gamma)$ and by Eq. (7.3),

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi_n^{(\gamma)}}{\mathrm{d}x^2} + V\left(x; \tfrac{\alpha}{\gamma}, \gamma L\right)\psi_n^{(\gamma)} = \frac{\gamma^{-1/2}}{\gamma^2}\left[-\frac{\hbar^2}{2m}\psi_n'' + V(\,\cdot\,; \alpha, L)\,\psi_n\right]\left(\frac{x}{\gamma}\right) = \frac{E_n}{\gamma^2}\,\psi_n^{(\gamma)}(x),$$

with the boundary condition moving with the walls and the norm preserved. The map is invertible, so

$$\boxed{E_n\left(\frac{\alpha}{\gamma}, \gamma L\right) = \frac{1}{\gamma^2} E_n(\alpha, L) \quad \text{for every } n.} \tag{7.4}$$

The exact equations agree: αL is unchanged and $\alpha \tanh(\alpha L/2) \to \alpha \tanh(\alpha L/2)/\gamma$, so the roots of Eqs. (2.9)–(2.11) scale as $k_j \to k_j/\gamma$, $\kappa_b \to \kappa_b/\gamma$. Numerically, $\max_n |\gamma^2 E_n(\text{after}) - E_n(\text{before})| \le 6 \times 10^{-17}$ for $\gamma = 0.7, 1.5, 2$. This is *homogeneous level scaling*, $E_n^c = qE_n^h$ with one $q = \gamma^{-2}$ for all n, in the language of Refs. [7, p. 2] and [6, p. 92].

7.4 Partition functions and heat capacities

Quantum: $Z_q(\beta; \frac{\alpha}{\gamma}, \gamma L) = \sum_n e^{-\beta E_n/\gamma^2} = Z_q(\beta/\gamma^2; \alpha, L)$. Classical: substitute $x = \gamma y$ and use Eq. (7.3) and $\lambda_{\mathrm{th}} \propto \beta^{1/2}$,

$$Z_{\mathrm{cl}}\left(\beta; \tfrac{\alpha}{\gamma}, \gamma L\right) = \frac{\gamma}{\lambda_{\mathrm{th}}(\beta)} \int e^{-(\beta/\gamma^2)V(y;\alpha,L)}\mathrm{d}y = Z_{\mathrm{cl}}\left(\frac{\beta}{\gamma^2}; \alpha, L\right).$$

Since $\beta^2\partial_\beta^2$ is unchanged by $\beta \to \beta/\gamma^2$,

$$C_v(T; \gamma) = C_v(\gamma^2 T; 1) \qquad \text{for both } C_v^{\mathrm{q}} \text{ and } C_v^{\mathrm{cl}}, \tag{7.5}$$

checked numerically to $2 \times 10^{-12} k_B$ (Fig. 3). The stroke slides every feature, the anomaly window included, from T^* to T^*/γ^2, without changing its shape or height. This is consistent with Sec. 1: the stroke keeps αL, and $k_B T/(\hbar^2\alpha^2/2m)$ is the same at $(T, \alpha/\gamma)$ as at $(\gamma^2 T, \alpha)$.

7.5 Why this matters for the engine: Q and W from C_v

This follows Levy and Gelbwaser-Klimovsky [6, pp. 92–96, Eqs. (4.15)–(4.18)] (Eq. (4.15) on p. 94, Eqs. (4.16)–(4.17) on p. 95, Eq. (4.18) on p. 96), in their notation: hot-stroke Hamiltonian H_h, cold-stroke Hamiltonian H_c, $E_n^c = qE_n^h$ (here $q = \gamma^{-2}$ for an expansion $\gamma > 1$). The Otto cycle is A $\to$ B (adiabatic $H_h \to H_c$) $\to$ C (cold bath T_c) $\to$ D (adiabatic $H_c \to H_h$) $\to$ A (hot bath T_h).

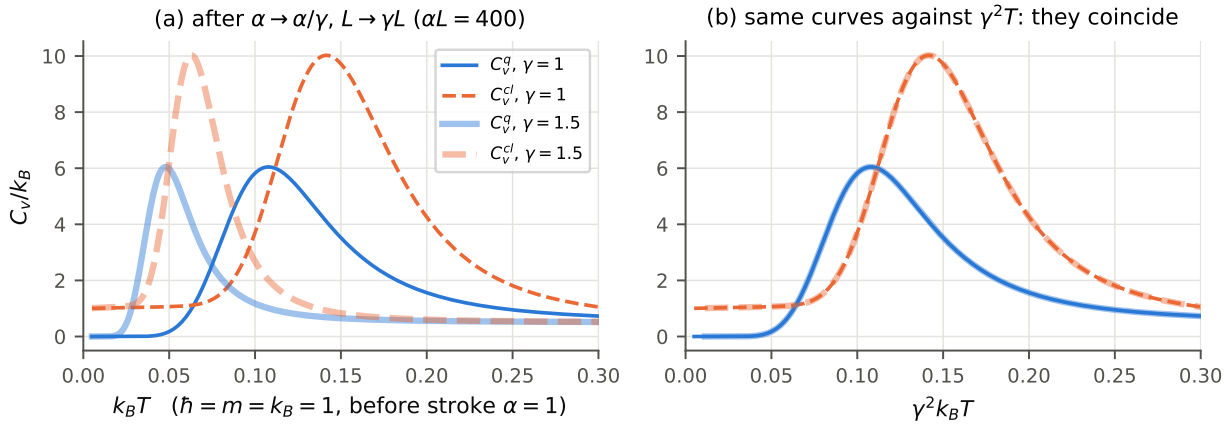


Figure 3: A scale-invariant stroke, $\alpha \to \alpha/\gamma$, $L \to \gamma L$ with $\gamma = 1.5$ ($\alpha L = 400$ kept). (a) The curves move to lower T. (b) Against $\gamma^2 T$ they coincide with the $\gamma = 1$ curves.

1. A is thermal at T_h for H_h: $p_n^A \propto e^{-E_n^h/k_B T_h}$.
2. A slow (quantum-adiabatic) stroke keeps the populations, and homogeneous scaling keeps the level order, so no level crossings occur.
3. C is thermal at T_c for H_c: $p_n^C \propto e^{-E_n^c/k_B T_c} = e^{-E_n^h/k_B (T_c/q)}$. After the stroke C $\to$ D the populations are unchanged, so D is *thermal for* H_h *at* T_c/q. Likewise B is thermal for H_c at qT_h.
4. Hence, with $U_h(T)$ the thermal energy of H_h and $U_c(T) = qU_h(T/q)$,

$$Q_h = \langle H_h \rangle_A - \langle H_h \rangle_D = U_h(T_h) - U_h(T_c/q) = \int_{T_c/q}^{T_h} C_v \, \mathrm{d}T, \tag{7.6}$$

$$Q_c = \langle H_c \rangle_C - \langle H_c \rangle_B = q\big[U_h(T_c/q) - U_h(T_h)\big] = -q \int_{T_c/q}^{T_h} C_v \, \mathrm{d}T, \tag{7.7}$$

$$W = -(Q_h + Q_c) = (q-1) \int_{T_c/q}^{T_h} C_v \, \mathrm{d}T, \qquad \eta = -\frac{W}{Q_h} = 1 - q, \tag{7.8}$$

where C_v is the heat capacity of H_h. An engine ($W < 0$) needs $T_c/T_h < q < 1$.

Gelbwaser-Klimovsky *et al.* [7, p. 2, Eq. (2)] state the same result with their q equal to our $1/q$, and note on the same page that it also follows from a fully classical treatment with the classical C_v. *Heuristic for the classical side:* in a slow 1D stroke the action $J = \oint p \, \mathrm{d}q$ is an adiabatic invariant [4, pp. 113–115, §4.6, Eq. (4.158)]. Under Eq. (7.3) an orbit maps as $x \to \gamma x$, $p \to p/\gamma$, which keeps J and multiplies H by γ^{-2}. So $E(J) \to E(J)/\gamma^2$ for every orbit, and a canonical distribution again maps to a canonical one at a scaled temperature.

Consequence. The efficiency is the same for both, $\eta = 1 - q$, but by Eq. (7.8) the work and heat are set by $\int C_v \, \mathrm{d}T$ alone. If $[T_c/q, T_h]$ lies inside a window where $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$, the quantum engine extracts more work at equal efficiency. The windows are narrow: at $\alpha L = 1000$, $[0.112, 0.203]$ ($\hbar^2\alpha^2/2m$), a ratio of 1.8, so $T_h/(T_c/q) \leq 1.8$.

8 What to double-check before presenting

1. **Every page and equation number in the references was checked against a copy:**
   - in `docs/refs`: Deffner *et al.* [1], Levy and Gelbwaser-Klimovsky [6] (book pages 87–98), and Gelbwaser-Klimovsky *et al.* [7] (the arXiv version, so its page numbers are arXiv pages);
   - free online copies: Cooper–Khare–Sukhatme (arXiv:hep-th/9405029v2, so arXiv pages), Tong's two sets of Cambridge lecture notes, and the DLMF equation numbers (`https://dlmf.nist.gov/1.8`, `/5.4`, `/5.5`, `/5.12`, `/7.2`, `/7.7`).

   Each cited result is also derived in this document, so no step relies on a reference alone.
2. **Links.** Every bibliography entry has a link, and every cited source is available online. Deffner *et al.* is open access. Cooper–Khare–Sukhatme, Gelbwaser-Klimovsky *et al.* and the Levy–Gelbwaser-Klimovsky chapter have free arXiv versions; for the chapter the arXiv pages differ from the book pages, and both are given. Tong's notes and the DLMF are free.
3. **Not cited, because no free copy was found to check pages against:** Flügge, *Practical Quantum Mechanics* (the source of Deffner's Pöschl–Teller row, cited here only through Deffner's Table I, their Ref. [55]), and Landau–Lifshitz, *Quantum Mechanics* (the classic textbook treatment of the levels of $-U_0/\cosh^2\alpha x$). Adding either to `docs/refs` would allow citing it with exact pages.
4. **Approximations:** terms of order $e^{-\alpha L}$ (Sec. 1) and the Poisson remainder R (exponentially small for $L \gtrsim \lambda_{\mathrm{th}}$) are dropped. Both are checked numerically against exact level sums.
5. **Heuristics are heuristics:** the equal-weight temperatures run 10–25% high and the peak-height estimate 5–15% high. They explain the ordering, not the exact numbers.
6. **Hard walls:** because of the wall term, $\int_0^\infty (C_v^{\mathrm{q}} - C_v^{\mathrm{cl}})\,\mathrm{d}T$ diverges here, so the zero-point sum rule that holds for smooth confining potentials does not apply.

A Direct check of ψ_k

With $u = \tanh(\alpha x)$, $u' = \alpha(1-u^2)$:

$$\begin{aligned}
\psi_k &= (\alpha u - ik)e^{ikx},\\
\psi_k' &= \big[\alpha^2(1-u^2) + ik\alpha u + k^2\big]e^{ikx},\\
\psi_k'' &= \big[-2\alpha^3 u(1-u^2) + 2ik\alpha^2(1-u^2) - k^2\alpha u + ik^3\big]e^{ikx}.
\end{aligned}$$

The Schrödinger equation is $-\psi'' - 2\alpha^2(1-u^2)\psi = k^2\psi$, since $\frac{2m}{\hbar^2}V = -2\alpha^2/\cosh^2(\alpha x) = -2\alpha^2(1-u^2)$ and $E = \hbar^2k^2/2m$. Substituting,

$$-\psi_k'' - 2\alpha^2(1-u^2)\psi_k = \big[k^2\alpha u - ik^3\big]e^{ikx} = k^2(\alpha u - ik)e^{ikx} = k^2\psi_k. \qquad \checkmark$$

B The Gaussian–Lorentzian integral

For $c > 0$ let $\mathcal{F}(b) = \int_0^\infty e^{-bk^2}/(k^2+c^2)\,\mathrm{d}k$. Then

$$\mathcal{F}'(b) - c^2\mathcal{F}(b) = -\int_0^\infty e^{-bk^2}\mathrm{d}k = -\frac{1}{2}\sqrt{\frac{\pi}{b}}, \qquad \text{so} \qquad \frac{\mathrm{d}}{\mathrm{d}b}\big(e^{-bc^2}\mathcal{F}\big) = -\frac{1}{2}\sqrt{\frac{\pi}{b}}\,e^{-bc^2}.$$

Integrate from b to ∞, where $e^{-bc^2}\mathcal{F} \to 0$, and substitute $s = u^2/c^2$:

$$\mathcal{F}(b) = e^{bc^2}\int_b^\infty \frac{1}{2}\sqrt{\frac{\pi}{s}}\,e^{-sc^2}\mathrm{d}s = \frac{\sqrt{\pi}}{c}\,e^{bc^2}\int_{c\sqrt{b}}^\infty e^{-u^2}\mathrm{d}u = \frac{\pi}{2c}\,e^{bc^2}\,\mathrm{erfc}\left(c\sqrt{b}\right),$$

which is DLMF Eq. (7.7.1) [5] after $k = ct$, $z = c\sqrt{b}$.

C The classical well integrals

$\alpha J_1 = \int_{-1}^1 e^{a(1-t^2)}\mathrm{d}t = 2e^a\int_0^1 e^{-at^2}\mathrm{d}t$, and with $u = t\sqrt{a}$, $\int_0^1 e^{-at^2}\mathrm{d}t = \frac{1}{\sqrt{a}}\int_0^{\sqrt{a}} e^{-u^2}\mathrm{d}u = \frac{\sqrt{\pi}}{2\sqrt{a}}\,\mathrm{erf}\,\sqrt{a}$. So $\alpha J_1 = \sqrt{\pi}\,e^a\,\mathrm{erf}\,\sqrt{a}/\sqrt{a}$. Differentiating, with $\mathrm{d}\,\mathrm{erf}\,\sqrt{a}/\mathrm{d}a = \frac{2}{\sqrt{\pi}}e^{-a}\frac{1}{2\sqrt{a}} = e^{-a}/\sqrt{\pi a}$,

$$\alpha J_2 = \frac{\mathrm{d}(\alpha J_1)}{\mathrm{d}a} = \sqrt{\pi}\,e^a\,\mathrm{erf}\,\sqrt{a}\left(a^{-1/2} - \tfrac{1}{2}a^{-3/2}\right) + \frac{1}{a} = \alpha J_1\left(1 - \frac{1}{2a}\right) + \frac{1}{a}.$$

Also $\mathrm{d}(\alpha I)/\mathrm{d}a = \alpha J_1$, consistent with Eq. (4.3): differentiating the series term by term gives $\sum_\ell 2^\ell a^{\ell-1}/(2\ell-1)!!$, which is the series of $2e^a\int_0^1 e^{-at^2}\mathrm{d}t$.

References

[1] S. Deffner, C. Jarzynski, and A. del Campo, "Classical and quantum shortcuts to adiabaticity for scale-invariant driving," Phys. Rev. X **4**, 021013 (2014). *Online (open access):* https://doi.org/10.1103/PhysRevX.4.021013. *Local copy:* `docs/refs/Classical and Quantum Shortcuts to Adiabaticity for Scale-Invariant Driving.pdf`. *Pages used:* 021013-2 (Eq. (3), the scale-invariant Hamiltonian); 021013-9 (Table I: the modified Pöschl–Teller row, citing Flügge as their Ref. [55], and the power-law row, which includes the infinite square well); 021013-14 (Appendix A, scale-invariant eigenfunctions).

[2] F. Cooper, A. Khare, and U. Sukhatme, "Supersymmetry and quantum mechanics," Phys. Rep. **251**, 267 (1995), https://doi.org/10.1016/0370-1573(94)00080-M. *Online (free), the version whose pages are cited:* arXiv:hep-th/9405029v2, https://arxiv.org/abs/hep-th/9405029v2. *Pages used:* 13–14 (Eqs. (3)–(7): factorization $H_1 = A^\dagger A$, $H_2 = AA^\dagger$, superpotential); 14 (Eqs. (8)–(11): partner eigenfunctions); 15 (Eqs. (12)–(14): equal spectra apart from the zero mode of A); 17 ("the same spectra except for the fact that H_2 has one fewer bound state"); 18 (Eqs. (32)–(33): $W = A\tanh x$; with a free-particle partner, the sech^2 well is reflectionless); 41–42 (Eqs. (150)–(153): $V_1 = n^2 - n(n+1)\mathrm{sech}^2 x$ is reflectionless, transmission amplitude); 77–78 (Eq. (312): "V_1 holds n bound states"; the case $n = 1$: $V_1 = -2\mathrm{sech}^2 x$, $E_1 = -1$, ground state $\propto \mathrm{sech}\,x$).

[3] D. Tong, *Statistical Physics*, University of Cambridge Part II lecture notes. *Online (free), the copy whose pages are cited:* https://www.damtp.cam.ac.uk/user/tong/statphys/sp.pdf. *Pages used:* 20 (§1.3.2, Eq. (1.27): $\Delta E^2 = k_B T^2 C_V$); 32 (§2.1, Eq. (2.1): classical partition function); 35 (Eq. (2.6): thermal de Broglie wavelength).

[4] D. Tong, *Classical Dynamics*, University of Cambridge Part II lecture notes. *Online (free), the copy whose pages are cited:* https://davidtong.org/pdfs/teaching/classical-dynamics/clas.pdf. *Pages used:* 113–115 (§4.6, adiabatic invariants; Eq. (4.158)).

[5] F. W. J. Olver *et al.* (eds.), *NIST Digital Library of Mathematical Functions*, https://dlmf.nist.gov. *Equations used, each at its permalink:* (1.8.15) Poisson's summation formula, https://dlmf.nist.gov/1.8.E15; (5.4.6) $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, https://dlmf.nist.gov/5.4.E6; (5.5.1) $\Gamma(z+1) = z\Gamma(z)$, https://dlmf.nist.gov/5.5.E1; (5.12.1) Euler's Beta integral, https://dlmf.nist.gov/5.12.E1; (7.2.1) definition of erf, https://dlmf.nist.gov/7.2.E1; (7.7.1) $\operatorname{erfc} z = \frac{2}{\pi} e^{-z^2} \int_0^\infty \frac{e^{-z^2 t^2}}{t^2+1} \mathrm{d}t$, https://dlmf.nist.gov/7.7.E1.

[6] A. Levy and D. Gelbwaser-Klimovsky, "Quantum features and signatures of quantum thermal machines," in *Thermodynamics in the Quantum Regime*, F. Binder, L. A. Correa, C. Gogolin, J. Anders, and G. Adesso (eds.), Fundamental Theories of Physics **195**, 87 (Springer, 2018), https://doi.org/10.1007/978-3-319-99046-0_4 (paywalled). *Local copy, the one whose (book) pages are cited:* `docs/refs/qvsclas book chapter (1).pdf`. *Online (free):* arXiv:1803.05586v1, https://arxiv.org/abs/1803.05586v1, with different page and equation numbers, given in brackets below. *Pages used, book [arXiv]:* 92 [p. 5] (homogeneous energy scaling $E_n^c = qE_n^h$); 94 [p. 6] (Eq. (4.15) [Eq. (15)]); 95 [pp. 6–7] (Eqs. (4.16)–(4.17) [Eqs. (16)–(17)]; the classical oscillator has $C_v = k_B$ by equipartition; Fig. 4.3 [Fig. 3], $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$ for the box); 96 [p. 7] (box discussion; Eq. (4.18) [Eq. (18)], $\eta = 1 - q$).

[7] D. Gelbwaser-Klimovsky, A. Bylinskii, D. Gangloff, R. Islam, A. Aspuru-Guzik, and V. Vuletić, "Single-atom heat machines enabled by energy quantization," Phys. Rev. Lett. **120**, 170601 (2018), https://doi.org/10.1103/PhysRevLett.120.170601 (paywalled). *Online (free), the version whose pages are cited:* arXiv:1705.11180v2, https://arxiv.org/abs/1705.11180v2. *Local copy:* `docs/refs/Single-atom heat machines enabled by energy quantization (2) (1).pdf`. *Pages used:* 2 (homogeneous scaling $E_{n,h} = qE_{n,c}$, whose q is our $1/q$; examples include changing the length of an infinite square well; Eq. (2); the same result from a classical treatment).